Microservices Project — Quick Reference

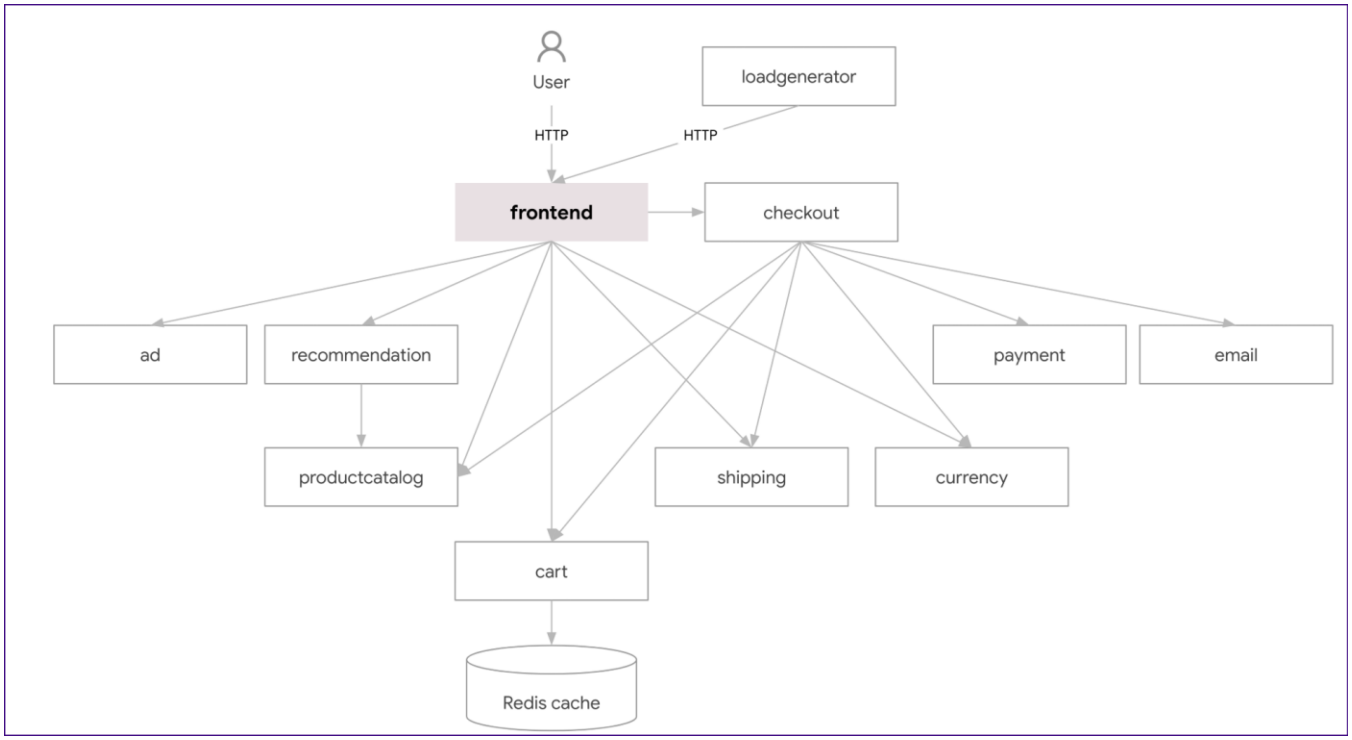
Online Boutique · Google Cloud Demo App

What is this project?

A fake e-commerce store (like Amazon/Flipkart) built with 12 small services that each do one job. Users can browse products, add to cart, and checkout. Built by Google to teach cloud-native DevOps.

Service Flow Diagram

Based on environment variables and service addresses defined in release/kubernetes-manifests.yaml



All 12 Services — What They Do

Service	Language	What it does
frontend	Go	Serves the website. Calls ALL other services. Only public-facing service.
checkoutservice	Go	Checkout brain. Calls cart → payment → shipping → email in sequence.
cartservice	C#	Stores your cart in Redis. Add item / get cart / empty cart.
productcatalogservice	Go	Returns product list from a JSON file. Search & get product by ID.
paymentservice	Node.js	Fake card payment. Validates card number (Luhn check) → returns transaction ID.
currencyservice	Node.js	Converts prices to user's currency. Uses real ECB rates. Highest traffic service.
shippingservice	Go	Returns shipping cost estimate. Simulates order dispatch.
emailservice	Python	Sends order confirmation email (mock — prints to logs, no real SMTP).
recommendationservice	Python	Suggests 'You may also like' products. Picks random items from catalog.
adservice	Java	Shows text ads on product pages. Matches ads to product keywords.
loadgenerator	Python	Not a real service — simulates user traffic (browse/add/checkout) for testing.
shoppingassistantservice	Python	AI chatbot powered by Google Gemini. Answers product questions in plain English.

Folder Structure

Folder	What's inside
/src/	Source code — one sub-folder per service
/kubernetes-manifests/	K8s YAML files — Deployment + Service for each microservice
/protos/	demo.proto — defines all gRPC APIs between services
/release/	Production-ready combined manifest + Ingress config
/docs/	Development guide, deployment steps, architecture notes
cloudbuild.yaml	CI/CD pipeline — builds Docker images on every commit
skaffold.yaml	Local dev tool — auto rebuild + redeploy when you save code

Key Concepts to Know

- gRPC — faster than REST, uses .proto files to define API contracts. All services use it.
- Redis — in-memory database. Only used by cartservice to store cart data.
- Polyglot — each service is written in a different language (Go, Python, Java, Node, C#).
- Kubernetes — each service runs in its own pod, managed by a Deployment + Service YAML.
- Frontend = only public entry point. All others are ClusterIP (internal only).